Team Control Number

16970

Problem Chosen

C

2025

HiMCM/MidMCM

Summary Sheet

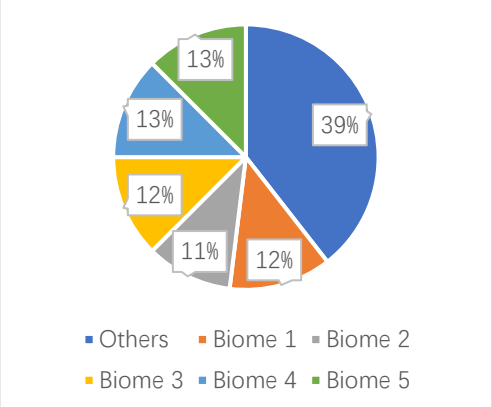
We were asked by the Council Obsessed with Maximizing Animal Prosperity (COMAP) to design a zoo with 1 square kilometer area. Our zoo, *Veridian Ark*, fulfills four primary purposes, which are wildlife conservation, scientific research, education, and recreation. The zoo mainly exhibits animals in five different biomes (tropical rainforest, savanna, desert / arid, temperate grassland, prairie, and boreal forest / taiga).

In our designing process, we first chose 30 common species for each of the 5 biomes. Next, we designed an evaluation model in Excel that aim for choosing species which have the highest scores of the combination of the four main purposes. Then, we input these species with relatively high rankings (mostly the highest 20 in overall score) into a Python program to get a more accurate selection of species. Finally, we designed the map of *Veridian Ark*, and gave each species a fixed area based on their size and scarcity.

Although the five biomes are the most essential part in our zoo, we also need to include:

- an aquarium
- a central square containing a lake with an in-lake restaurant
- a parking lot
- staff area

The purpose of the aquarium is to increase the biodiversity, and the rest of the designs focuses on cultivating *recreation*, which further increases *visitor's experience*. The figure below showcases the percentages of different sections in the zoo.



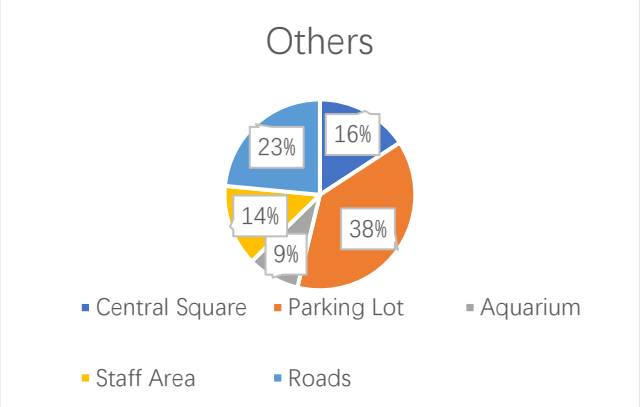


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1. Introduction

1.1 Restatement of the Problem

Our team had been asked by the Council Obsessed with Maximizing animal Prosperity (COMAP) to design a one square kilometer zoo. The four primary purposes of constructing this zoo are *wildlife conservation*, *scientific research*, *education*, *recreation*. Below is our interpretation to a zoo that can satisfy these requirements:

- Q: How can a zoo achieve the four primary purposes?
  A: Wildlife conservation can be achieved by ensuring the safety of the species; scientific research can be achieved by involving species with high scarcity; education can be achieved by special information about some species (i.e., *Falco peregrinus* is the fast bird); recreation can be achieved by involving relaxing activities like food stands, lake tour, safari ride, and interaction with animals.
- Q: What do you need to consider for the animals?
  A: We need to consider the species' living conditions, predatory relationships, and cohabitational relationships.
- Q: What do you need to consider for the employees?
  A: We can give employees a living area in the zoo and some resting places for employees in each part of the map.
- Q: What do you need to consider for the visitors?
  A: For visitors' safety, every aggressive species is being kept individually in cages. For visitors' amenities, stands and restaurants are designed in the zoo.

1.2 Assumptions & Justifications

Some of the conditions we supposed in our zoo model is not stated in the problem, though they are necessary in our model. Thy are being listed below.

Table 1.2.1 Assumptions & Justifications

Assumption 1 The peak pedestrian volume per day is about 30,000 persons, about 60% of all visitors go to the zoo by car, and each car can averagely take three persons.	**Justification 1** We researched that the pedestrian volume of a local zoo with normal size is at most 30,000 persons per day. The rest of the assumption also comes from researches.

Assumption 2 Our zoo can contain about 100 different animal species (except ones in the aquarium).	**Justification 2** We calculated that the actual place for animals is about 250,000 square meters (except for the aquarium). We researched that an area this big can contain 100 species.
Assumption 3 Our zoo has one aquarium and 5 areas containing 5 different biomes of species.	**Justification 3** To achieve wildlife conservation, we should provide the animals with appropriate living environments in a zoo. Therefore, exhibits should simulate the living conditions of a biome. It is reasonable to keep different species of the same biome together, as this maximizes the use of limited space while ensuring the animals' well-being.
Assumption 4 We will include souvenir shops, restaurants, washrooms in the zoo, and a lake in the zoo.	**Justification 4** Recreation is included in the four primary purposes, and these settings can help achieve recreation.
Assumption 5 In our model, while guaranteeing the well-being of the animals, we will use as few resources as possible.	**Justification 5** Though there are significant resources available, the resources are not unlimited. Hence, the fewer resources used, the better.
Assumption 6 The ratio between student visitors and adult visitors is 2:3.	**Justification 6** Student visitors are often accompanied by adults, so although zoos are mainly focused on students, the percentage of adult visitors will be slightly higher.
Assumption 7 The shape of the zoo is a square.	**Justification 7** A square-shaped map will be more explicit to visitors, because most of the lines are either horizontal or vertical, making it less possible for visitors to get lost.

2. Model Presentation

2.1 Evaluation Model

We firstly interpreted and separated the four primary purposes (*wildlife conservation*, *scientific research*, *education*, *recreation*) of zoos:

- *Scientific research* is only based on the scarcity and diversity of the species. Diversity is not considered in our evaluation model, as we already assumed that we would include all types of animals (i.e., mammals, fish, etc.), and there are about 20 species chosen in each biome. Based on the data from the International Union for Conservation of Nature (IUCN), the scarcity of species is divided into five main levels: critically endangered (CR), endangered (EN), vulnerable (VU), near threatened (NT), and least concerned (LC). In our model, a higher scarcity level is reflected by a higher *scientific research* value.
- *Wildlife conservation* is mainly based on the feeding costs of the species. The species with lower feeding costs have a higher rank in our model. Noticeably, if the scarcity of the species is high, we will slightly increase the score. For instance, if the score of *scientific research* is 0.75, which represents a VN species, we will increase the *wildlife conservation* score by 0.01.
- The evaluation of *education* and *recreation* is divided into students and adults. The purpose of a student visiting and an adult visiting is different: the weight of education against recreation is significantly higher for students than for adults.
- To better show the overall result of education and evaluation, we use *visitor experience* to showcase the combination of *student education*, *student recreation*, *adult education*, and *adult recreation*.

Now, the whole result is affected by three factors: visitor experience, scientific research, and wildlife conservation. *Visitor experience* is constituted by four smaller purposes, so we consider this first. As claimed in **Assumption 6** of **1.2**, we supposed that the percentage of student visitors and adult visitors is 2:3. Some student visitors come to the zoo because of field trips of schools or parents' wish to make students learn more about animals, so we assume that the weight of *student education* and *student recreation* is 3:2. Therefore, the total percentage of *student education* in consideration of *visitor experience* is:

$$\frac{2}{5} \times \frac{3}{5} = \frac{6}{25}$$

Similarly, the percentage of *student recreation* is:

$$\frac{2}{5} \times \frac{2}{5} = \frac{4}{25}$$

After that, we consider the situation for adults. This is much different from considering students: adults visit mostly to relax in leisure time, and they already know quite a lot about animals compared to students. Accordingly, the weight of *adult recreation* would be significantly higher than *adult recreation*: we gave a ratio of 9:1. As a result, the total percentages of *recreation* and *education* for adults are

$$\frac{3}{5} \times \frac{9}{10} = \frac{27}{50}$$

and

$$\frac{3}{5} \times \frac{1}{10} = \frac{3}{50}$$

respectively.

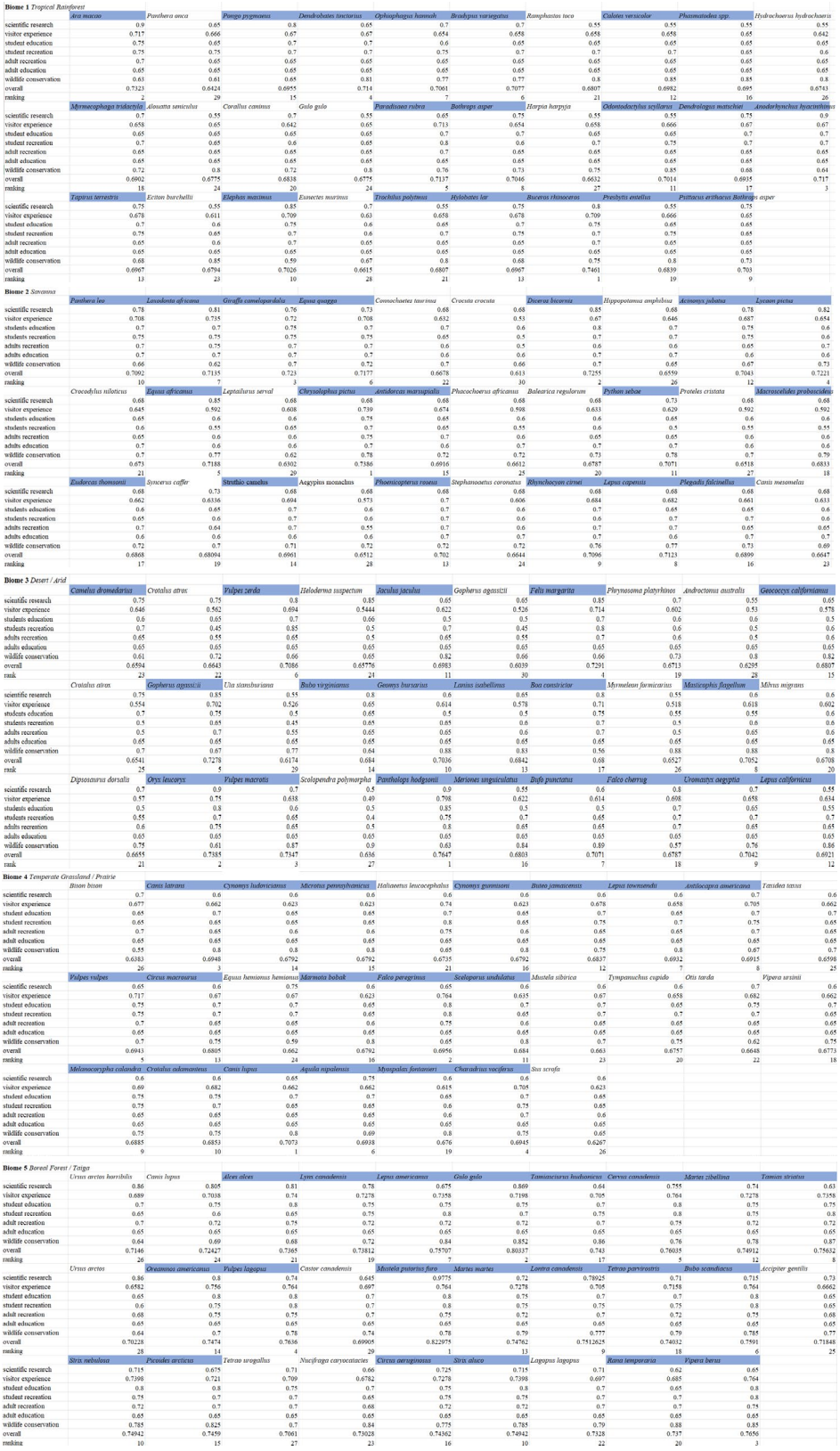
After the factors of *visitor experience* is established, we tried to determine the weight of each data overall. Since visitor experience is the combination of two purposes, we gave it the highest weight: 0.4. Next, as the most basic and most important purpose of the zoo, protecting animals and making it health obviously belongs to the category of *wildlife conservation.* Consequently, we gave *wildlife conservation* a higher weight than scientific research. We made the two categories weigh 0.35 and 0.25, respectively. Hence, the algorithm of the overall ranking score is:

$$0.4 \times \textit{visitor experience} + 0.35 \times \textit{wildlife conservation} +$$

$$0.25 \times \textit{scientific research}$$

This ranking score can comprehensively show whether an animal species is suitable to be in a zoo.

All the calculations above were done in Excel. Results are shown in the Excel graph below (The grids marked blue are our final decision on animal selection. It is unnecessary to be examined in this section). After getting the final score of each species, we gave them a ranking. The ranking is the preliminary results of animal choosing: the species with ranking 1~20 was likely to be finally chosen. However, the final choice is done by a Python program. This will be elaborated in detail in **2.2**.



Biome 1 *Tropical Rainforest*

	Ara macao	*Panthera onca*	*Pongo pygmaeus*	*Dendrobates tinctorius*	*Ophiophagus hannah*	*Bradypus variegatus*	*Ramphastos toco*	*Calotes versicolor*	*Phasmatodea spp.*	*Hydrochoerus hydrochaeris*
scientific research	0.9	0.65	0.8	0.65	0.7	0.7	0.55	0.55	0.55	0.55
visitor experience	0.717	0.666	0.67	0.67	0.654	0.658	0.658	0.658	0.65	0.642
student education	0.75	0.65	0.7	0.7	0.6	0.65	0.65	0.65	0.65	0.65
student recreation	0.75	0.75	0.7	0.7	0.75	0.7	0.7	0.7	0.65	0.6
adult recreation	0.7	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
adult education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.63	0.61	0.65	0.81	0.77	0.77	0.8	0.85	0.85	0.8
overall	0.7323	0.6424	0.6955	0.714	0.7061	0.7077	0.6807	0.6982	0.695	0.6743
ranking	2	29	15	4	7	6	21	12	16	26

	Myrmecophaga tridactyla	*Alouatta seniculus*	*Corallus caninus*	*Gulo gulo*	*Paradisaea rubra*	*Bothrops asper*	*Harpia harpyja*	*Odontodactylus scyllarus*	*Dendrolagus matschiei*	*Anodorhynchus hyacinthinus*
scientific research	0.7	0.55	0.7	0.55	0.65	0.75	0.55	0.55	0.75	0.9
visitor experience	0.658	0.65	0.642	0.65	0.713	0.654	0.658	0.666	0.67	0.67
student education	0.65	0.65	0.65	0.65	0.7	0.7	0.65	0.65	0.7	0.7
student recreation	0.7	0.65	0.6	0.65	0.8	0.8	0.7	0.75	0.7	0.7
adult recreation	0.65	0.65	0.65	0.65	0.7	0.65	0.65	0.65	0.65	0.65
adult education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.72	0.8	0.72	0.8	0.76	0.73	0.75	0.85	0.68	0.64
overall	0.6902	0.6775	0.6838	0.6775	0.7137	0.7046	0.6632	0.7014	0.6935	0.717
ranking	18	24	20	24	5	8	27	11	17	3

	Tapirus terrestris	*Eciton burchellii*	*Elephas maximus*	*Eunectes murinus*	*Trichilus polytmus*	*Hylobates lar*	*Buceros rhinoceros*	*Presbytis entellus*	*Psittacus erithacus*	*Bothriechis aspe*
scientific research	0.75	0.55	0.85	0.7	0.55	0.75	0.8	0.55	0.75	
visitor experience	0.678	0.611	0.709	0.63	0.658	0.678	0.709	0.666	0.65	
student education	0.7	0.6	0.75	0.6	0.65	0.7	0.75	0.65	0.65	
student recreation	0.75	0.65	0.7	0.6	0.7	0.75	0.7	0.75	0.65	
adult recreation	0.65	0.6	0.7	0.65	0.65	0.65	0.7	0.65	0.65	
adult education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	
wildlife conservation	0.68	0.85	0.59	0.67	0.8	0.68	0.75	0.8	0.73	
overall	0.6967	0.6794	0.7026	0.6615	0.6807	0.6967	0.7461	0.6839	0.703	
ranking	13	23	10	28	21	13	1	19	9	

Biome 2 *Savanna*

	Panthera leo	*Loxodonta africana*	*Giraffa camelopardalis*	*Equus quagga*	*Connochaetes taurinus*	*Crocuta crocuta*	*Diceros bicornis*	*Hippopotamus amphibius*	*Acinonyx jubatus*	*Lycaon pictus*
scientific research	0.78	0.81	0.76	0.73	0.68	0.68	0.85	0.68	0.78	0.82
visitor experience	0.708	0.735	0.72	0.708	0.632	0.53	0.67	0.646	0.687	0.654
students education	0.7	0.7	0.75	0.7	0.7	0.6	0.8	0.7	0.75	0.6
students recreation	0.75	0.75	0.75	0.75	0.65	0.5	0.7	0.7	0.75	0.6
adults recreation	0.7	0.75	0.7	0.7	0.6	0.5	0.6	0.6	0.65	0.7
adults education	0.7	0.7	0.7	0.7	0.6	0.6	0.7	0.7	0.6	0.6
wildlife conservation	0.66	0.62	0.7	0.72	0.7	0.66	0.7	0.65	0.67	0.73
overall	0.7092	0.7135	0.723	0.7177	0.6678	0.613	0.7255	0.6559	0.7043	0.7221
ranking	10	7	3	6	22	30	2	26	12	4

	Crocodylus niloticus	*Equus africanus*	*Leptailurus serval*	*Chrysolophus pictus*	*Antidorcas marsupialis*	*Phacochoerus africanus*	*Balearica regulorum*	*Python sebae*	*Proteles cristata*	*Macroscelides proboscideus*
scientific research	0.68	0.85	0.68	0.68	0.68	0.68	0.68	0.73	0.68	0.68
visitor experience	0.645	0.592	0.608	0.739	0.674	0.598	0.633	0.629	0.592	0.592
students education	0.65	0.6	0.6	0.75	0.65	0.6	0.6	0.65	0.6	0.6
students recreation	0.6	0.55	0.65	0.7	0.65	0.55	0.6	0.5	0.55	0.55
adults recreation	0.65	0.6	0.6	0.75	0.7	0.6	0.65	0.65	0.6	0.6
adults education	0.7	0.6	0.6	0.7	0.6	0.7	0.7	0.7	0.6	0.6
wildlife conservation	0.7	0.77	0.62	0.78	0.72	0.72	0.73	0.78	0.7	0.79
overall	0.673	0.7188	0.6302	0.7386	0.6916	0.6612	0.6787	0.7071	0.6518	0.6833
ranking	21	5	29	1	15	25	20	11	27	18

	Eudorcas thomsonii	*Syncerus caffer*	*Struthio camelus*	*Aegypius monachus*	*Phoeniconeterus roseus*	*Stephanoaetus coronatus*	*Rhynchocyon cirnei*	*Lepus capensis*	*Plegadis falcinellus*	*Canis mesomelas*
scientific research	0.68	0.73	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68
visitor experience	0.662	0.6336	0.694	0.573	0.7	0.606	0.684	0.682	0.661	0.633
students education	0.6	0.65	0.7	0.6	0.7	0.6	0.7	0.65	0.65	0.6
students recreation	0.65	0.6	0.7	0.6	0.7	0.6	0.6	0.7	0.7	0.6
adults recreation	0.7	0.64	0.7	0.55	0.7	0.6	0.7	0.7	0.65	0.65
adults education	0.6	0.6	0.6	0.6	0.7	0.7	0.7	0.6	0.7	0.7
wildlife conservation	0.72	0.7	0.71	0.72	0.72	0.72	0.76	0.77	0.73	0.69
overall	0.6868	0.68094	0.6961	0.6512	0.702	0.6644	0.7096	0.7123	0.6899	0.6647
ranking	17	19	14	28	13	24	9	8	16	23

Biome 3 *Desert / Arid*

	Camelus dromedarius	*Crotalus atrox*	*Vulpes zerda*	*Heloderma suspectum*	*Jaculus jaculus*	*Gopherus agassizii*	*Felis margarita*	*Phrynosoma platyrhinos*	*Andrectonus australis*	*Geococcyx californianus*
scientific research	0.75	0.75	0.8	0.85	0.65	0.65	0.85	0.7	0.55	0.65
visitor experience	0.646	0.562	0.694	0.5444	0.622	0.526	0.714	0.602	0.53	0.578
students education	0.6	0.65	0.7	0.66	0.5	0.5	0.7	0.6	0.6	0.5
students recreation	0.7	0.45	0.85	0.5	0.7	0.45	0.8	0.6	0.5	0.6
adults recreation	0.65	0.55	0.65	0.5	0.65	0.55	0.7	0.6	0.5	0.6
adults education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.61	0.72	0.66	0.65	0.82	0.66	0.66	0.73	0.8	0.82
overall	0.6594	0.6643	0.7086	0.6577	0.6983	0.6039	0.7291	0.6713	0.6295	0.6807
rank	23	22	6	24	11	30	4	19	28	15

	Crotalus atrox	*Gopherus agassizii*	*Uta stansburiana*	*Bubo virginianus*	*Geomys bursarius*	*Lanius isabellinus*	*Boa constrictor*	*Myrmelaon formicarius*	*Masticophis flagellum*	*Milvus migrans*
scientific research	0.75	0.55	0.55	0.8	0.6	0.65	0.8	0.55	0.6	0.6
visitor experience	0.554	0.702	0.526	0.65	0.614	0.578	0.71	0.518	0.618	0.602
students education	0.7	0.75	0.5	0.65	0.5	0.5	0.75	0.55	0.55	0.6
students recreation	0.5	0.65	0.45	0.65	0.65	0.6	0.7	0.5	0.6	0.6
adults recreation	0.5	0.7	0.55	0.65	0.65	0.6	0.7	0.5	0.65	0.6
adults education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.7	0.67	0.77	0.64	0.88	0.83	0.56	0.88	0.88	0.8
overall	0.6541	0.7278	0.6174	0.684	0.7036	0.6842	0.68	0.6527	0.7052	0.6708
rank	25	5	29	14	10	13	17	26	8	20

	Dipsosaurus dorsalis	*Oryx leucoryx*	*Vulpes macrotis*	*Scolopendra polymorpha*	*Panthologa hodgsoni*	*Marisones unguiculatus*	*Bufo punctatus*	*Falco cherrug*	*Uromastyx aegyptia*	*Lepus californicus*
scientific research	0.7	0.9	0.7	0.5	0.9	0.55	0.6	0.8	0.7	0.55
visitor experience	0.57	0.75	0.638	0.49	0.798	0.622	0.614	0.698	0.658	0.634
students education	0.5	0.8	0.6	0.5	0.85	0.5	0.5	0.7	0.65	0.55
students recreation	0.55	0.7	0.65	0.4	0.75	0.7	0.65	0.7	0.7	0.7
adults recreation	0.6	0.75	0.65	0.5	0.8	0.65	0.65	0.7	0.65	0.65
adults education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.75	0.61	0.87	0.9	0.63	0.84	0.89	0.57	0.76	0.86
overall	0.6655	0.7385	0.7547	0.616	0.7647	0.6803	0.7071	0.6767	0.7042	0.6921
rank	21	2	3	27	1	16	7	18	9	12

Biome 4 *Temperate Grassland / Prairie*

	Bison bison	*Canis latrans*	*Cynomys ludovicianus*	*Microtus pennsylvanicus*	*Haliaeetus leucocephalus*	*Cynomys gunnisoni*	*Buteo jamaicensis*	*Lepus townsendii*	*Antilocapra americana*	*Taxidea taxus*
scientific research	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.7	0.6
visitor experience	0.677	0.662	0.623	0.674	0.74	0.623	0.678	0.658	0.705	0.662
student education	0.65	0.7	0.65	0.65	0.7	0.65	0.7	0.65	0.7	0.7
student recreation	0.65	0.65	0.65	0.65	0.8	0.65	0.75	0.7	0.75	0.65
adult recreation	0.7	0.65	0.6	0.6	0.75	0.6	0.65	0.65	0.7	0.65
adult education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.55	0.8	0.8	0.8	0.65	0.8	0.75	0.8	0.67	0.7
overall	0.6383	0.6948	0.6792	0.6792	0.6735	0.6792	0.6837	0.6932	0.6915	0.6598
ranking	26	3	14	15	21	16	12	7	8	25

	Vulpes vulpes	*Circus macrourus*	*Equus hemionus hemionus*	*Marmota bobak*	*Falco peregrinus*	*Scelopurus undulatus*	*Mustela sibirica*	*Tympanuchus cupido*	*Otis tarda*	*Vipera ursinii*
scientific research	0.65	0.6	0.75	0.6	0.65	0.6	0.6	0.6	0.7	0.6
visitor experience	0.717	0.67	0.67	0.623	0.764	0.635	0.67	0.658	0.682	0.662
student education	0.75	0.7	0.7	0.65	0.8	0.7	0.7	0.65	0.75	0.7
student recreation	0.75	0.7	0.7	0.7	0.65	0.8	0.65	0.7	0.7	0.65
adult recreation	0.7	0.65	0.65	0.6	0.75	0.6	0.65	0.65	0.65	0.65
adult education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.7	0.75	0.59	0.8	0.65	0.8	0.7	0.75	0.62	0.75
overall	0.6943	0.6805	0.662	0.6792	0.6956	0.684	0.663	0.6757	0.6648	0.6773
ranking	5	13	24	16	2	11	23	20	22	18

	Melanocorypha calandra	*Crotalus adamanteus*	*Canis lupus*	*Aquila nipalensis*	*Myospalax fontanieri*	*Charadrius vociferus*	*Sus scrofa*
scientific research	0.6	0.6	0.65	0.75	0.6	0.6	
visitor experience	0.69	0.682	0.662	0.662	0.615	0.705	0.623
student education	0.75	0.75	0.7	0.7	0.65	0.7	0.65
student recreation	0.75	0.7	0.65	0.65	0.6	0.75	0.65
adult recreation	0.65	0.65	0.65	0.65	0.6	0.7	0.6
adult education	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.75	0.75	0.8	0.69	0.8	0.75	0.65
overall	0.6885	0.6853	0.7073	0.6938	0.676	0.6945	0.6267
ranking	9	10	1	6	19	4	26

Biome 5 *Boreal Forest / Taiga*

	Ursus arctos horribilis	*Canis lupus*	*Alces alces*	*Lynx canadensis*	*Lepus americanus*	*Gulo gulo*	*Tamiasciurus hudsonicus*	*Corvus canadensis*	*Martes zibellina*	*Tamias striatus*
scientific research	0.86	0.805	0.81	0.78	0.675	0.869	0.64	0.755	0.74	0.63
visitor experience	0.689	0.7038	0.74	0.7278	0.7358	0.7198	0.705	0.764	0.7278	0.7358
student education	0.7	0.75	0.8	0.75	0.75	0.75	0.7	0.8	0.75	0.75
student recreation	0.65	0.6	0.65	0.75	0.8	0.7	0.75	0.8	0.75	0.8
adult recreation	0.7	0.72	0.75	0.72	0.72	0.72	0.7	0.75	0.72	0.72
adult education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.64	0.69	0.68	0.72	0.84	0.852	0.86	0.76	0.78	0.87
overall	0.7146	0.72427	0.7365	0.73812	0.75707	0.80337	0.743	0.76035	0.74912	0.75632
ranking	26	24	21	19	7	2	17	5	12	8

	Ursus arctos	*Dreamnos americanus*	*Vulpes lagopus*	*Castor canadensis*	*Mustela putorius furo*	*Martes martes*	*Contra canadensis*	*Tetrao parvirostris*	*Bubo scandiacus*	*Accipiter gentilis*
scientific research	0.86	0.8	0.74	0.645	0.9775	0.72	0.78925	0.71	0.715	0.73
visitor experience	0.6582	0.756	0.764	0.697	0.764	0.7278	0.705	0.7158	0.764	0.6662
student education	0.65	0.8	0.8	0.7	0.8	0.75	0.7	0.7	0.8	0.65
student recreation	0.6	0.75	0.8	0.7	0.8	0.75	0.75	0.75	0.8	0.65
adult recreation	0.68	0.75	0.75	0.7	0.75	0.72	0.7	0.72	0.75	0.68
adult education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.64	0.7	0.78	0.74	0.78	0.79	0.777	0.79	0.785	0.77
overall	0.70228	0.7474	0.7636	0.66905	0.822975	0.74782	0.7512625	0.74032	0.7501	0.71848
ranking	28	14	4	29	1	13	9	18	6	25

	Strix nebulosa	*Picoides arcticus*	*Tetrao urogallus*	*Nucifraga caryocatactes*	*Circus aeruginosus*	*Strix aluco*	*Lagopus lagopus*	*Rana temporaria*	*Vipera berus*
scientific research	0.715	0.675	0.71	0.66	0.725	0.715	0.71	0.62	0.65
visitor experience	0.7398	0.721	0.709	0.6782	0.7278	0.7398	0.697	0.685	0.764
student education	0.8	0.8	0.75	0.7	0.75	0.8	0.7	0.65	0.8
student recreation	0.75	0.7	0.7	0.65	0.75	0.75	0.7	0.7	0.8
adult recreation	0.72	0.7	0.7	0.68	0.72	0.72	0.7	0.7	0.75
adult education	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
wildlife conservation	0.785	0.825	0.7	0.84	0.775	0.785	0.79	0.88	0.85
overall	0.74942	0.7459	0.7061	0.73028	0.74362	0.74942	0.7328	0.737	0.7656
ranking	10	15	27	23	16	10	22	20	3

Figure 2.1.1 Overall Data Derived from Evaluation Model

To show our steps more clearly, in the following text, we will use an example, the species *Pongo pygmaeus*, to illustrate how our evaluation model is constructed. The data of the species is shown in **Figure 2.1.2**.

	A	B
1	**Biome 1** *Tropical Rainforest*	
2		*Pongo pygmaeus*
3	scientific research	0.8
4	visitor experience	0.67
5	student education	0.7
6	student recreation	0.7
7	adult recreation	0.65
8	adult education	0.65
9	wildlife conservation	0.65
10	overall	0.6955
11	ranking	15

Figure 2.1.2 Data of *Pongo pygmaeus*

According to data from IUCN, the scarcity of this species is CR, so we gave it 0.8 on *scientific research*, which is a high value, since the value of *scientific research* in our animal depends on the scarcity of the species. *Student education* and *student recreation* is our subjective score after learning about the species. We thought that the appearance of the species is somewhat attractive, and there are indeed some interesting knowledge of the species, so we gave two 0.7's. The value of *adult recreation* and *adult education* is mostly based on the value in B5 and B6, since adults' value won't differ so much from students' value. However, for adults we reduced the dispersion of the value, making the variance of all species' value smaller, since adults' feelings are not as easily fluctuated as students', and adults probably have already known something about the species before visiting the zoo. Consequently, we gave two 0.65's. Then, because of the feeding cost of *Pongo pygmaeus* is relatively high, we gave 0.6 as a primitive value and then added 0.05 to it because of its high scarcity. Next, we can calculate the value of *visitor experience*. Typing the formula shown in **Figure 2.1.3**(a) into Excel, we could know that B6 should be 0.67. Finally, we can know the *overall* value using the formula shown in **Figure 2.1.3**(b), and give it a ranking based on the *overall* value of other species.

	A	B
1	**Biome 1** *Tropical Rainforest*	
2		*Pongo pygmaeus*
3	scientific research	0.8
4	visitor experience	=6/25*B5+4/25*B6+27/50*B7+3/50*B8
5	student education	0.7
6	student recreation	0.7
7	adult recreation	0.65
8	adult education	0.65
9	wildlife conservation	0.65
10	overall	0.6955
11	ranking	15

(a)

	A	B
1	**Biome 1** *Tropical Rainforest*	
2		*Pongo pygmaeus*
3	scientific research	0.8
4	visitor experience	0.67
5	student education	0.7
6	student recreation	0.7
7	adult recreation	0.65
8	adult education	0.65
9	wildlife conservation	0.65
10	overall	=0.4*B4+0.35*B9+0.25*B3
11	ranking	

(b)

Figure 2.1.3 Formulae in Excel of the Evaluation Model

2.2 Animal Selection

150 animal species are chosen into the biome evaluation model (30 animal species each), however, as mentioned in **1.2**, only about 100 animal species will be contained in the zoo.

Hence, an animal selecting system is needed, using the overall ranking score that is calculated in 2.1.

```
rankList = [0.7323, 0.6424, 0.6955, 0.714, 0.7061, 0.7077, 0.6807, 0.6982, \
            0.695, 0.6743, 0.6902, 0.6775, 0.6838, 0.6775, 0.7137, 0.7046, \
            0.6632, 0.7014, 0.6935, 0.717, 0.6967, 0.6794, 0.7026, 0.6615, \
            0.6807, 0.6967, 0.7461, 0.6839, 0.703]
IdList = []
sizeList = [1, 3, 3, 1, 2, 2, 1, 1, 1, 2, 2, 2, 2, 3, 1, 1, 2, 1, 2, 1, 2, 1, 3,
          2, 2, 2, 2, 2, 1]
for m in range(len(rankList)):
    IdList.append(m + 1)
for a in range(len(rankList) - 1):
    flag = 0
    for i in range(len(rankList) - 1 - a):
        if rankList[i] < rankList[i + 1]:
            rankList[i], rankList[i + 1] = rankList[i + 1], rankList[i]
            IdList[i], IdList[i + 1] = IdList[i + 1], IdList[i]
            sizeList[i], sizeList[i + 1] = sizeList[i + 1], sizeList[i]
            flag = 1
    if flag == 0:
        break
print(rankList)
print(IdList)
sumchoose = 0
chosenList = []
for i in range(20):
    chosenList.append(IdList[i])
    sumchoose += sizeList[i]
print(chosenList)
if 1.6 <= sumchoose / 20 <= 2.4:
    print('OK:', chosenList, sumchoose / 20)
```

Figure 2.2.1 Python Program Used in Animal Selection

As shown in **Figure 2.2.1**, we wrote a Python program; it will first sort the 30 overall ranks of the animal species in the input. A well-designed biome section in a zoo should include animal species with every size, though animal species in some specific biome could be relatively larger (i.e., animals in the Temperate Grassland / Prairie Biome are larger than other biome). Containing animal species of every size can more effectively show the biodiversity, and also give visitor better visual effect. We developed a ranking system only using 1, 2, and 3 to represent the size of an

animal species (1 represents small-sized species, 3 represents big-sized ones, and 2 represents ones that are middle-sized). If the arithmetic mean of the 20 chosen animal species' size rank is between 1.6~2.4 - which means that animal species of every size is selected, also with a moderate average - then it will be an ideal selection and will be printed in the program. Else, the program will substitute the 20th animal (after sorted) with the 21th animal (after sorted), then calculate the mean of animal species' size rank again. If the result is still too high or too low, the selection will be rearranged manually, until the mean satisfies:

$$1.6 <= \text{mean} <= 2.4$$

In this process, animal species with better appearance or higher conservation value would be more likely to be chosen, while chosen species with poor appearance or lower conservation value would be more likely to be substituted.

If most of the animals are big-sized in a biome, since the available land for each biome is limited, there could only be 18-19 animal species chosen in that biome (vice versa). It's also possible to have an animal species is small in size itself but needs a big living environment, while there's still enough space for this species, it can still be selected. However, this kind of animal could also be substituted if there's not enough space for it. Similarly, if an animal species needs very special living condition that is difficult to achieve in a normal zoological park, it could be substituted too.

In some case, an animal species could have their predator species in the same biome, and these two species are both selected. In order to ensure the animals' well-being, the selection will also be adjusted based on the output of the program, the animal species with higher ranking score will be chosen and the other one will probably be substituted. However, if the two animal species with natural enemy relationship both have very high conservation value, they can still be selected, but need to live apart with a greater distance in the zoo (for instance, on the two corners of the biome section, ensuring the living safety of both species).

If two animal species in a biome have symbiotic relationship, they could be both selected even though one of them have relatively lower ranking score.

After this selection, about 100 species are left. They are our final choice of animals and are marked blue in **Figure 2.1.1**.

2.3 Mapping (note: all of our maps are drawn on https://www.drawio.com/)

According to **Assumption 7** in **1.2**, our zoo, *Veridian Ark*, was designed as a square shape. Next, we will organize the different parts of our zoo.

We have previously stated that our zoo will include five zones, each simulating

the climate and living conditions of a specific biome. Below are the five biomes:

- **Biome 1**: Tropical Rainforest Biome

  Climate description: Consistently warm and humid year-round with heavy rainfall.

- **Biome 2**: Savanna Biome

  Climate description: Hot year-round with a distinct dry season and a rainy season.

- **Biome 3**: Desert / Arid Biome

  Climate description: Very low rainfall, extreme temperature shifts between day and night.

- **Biome 4**: Temperate Grassland / Prairie Biome

  Climate description: Hot summers, cold winters, moderate rainfall.

- **Biome 5**: Boreal Forest / Taiga Biome

  Climate description: Long, very cold winters; short, cool summers.

Biome 1, **biome 3**, **biome 4**, and **biome 5** are designed as exhibitions of their corresponding animals, while **biome 2** features a more free-roaming environment for the animals, allowing visitors to experience a safari ride amongst the vast grassland. We considered this specific design of great *recreation* value.

We further reasoned that a proper zoo should also include: 1) a parking lot, 2) a square for major activities, 3) restaurants & stands for eating, and 4) roads wide enough for visitors to move around freely. Together with the area of our five biomes and an aquarium, our total area should be no more than 1 square kilometer. Below are the areas each part should occupy:

1) A parking lot

In order to calculate the approximate area of the parking lot, we will first need to estimate the number of cars. The instantaneous vehicle count (IVC) can be calculated by the following formula:

$$IVC = \frac{v_{total} \times p_{car}}{v_{per\ car}}$$

where v_{total} is the total number of visitors at an instant, p_{car} is the possibility an individual visitor will come to the zoo by car, and $v_{per\ car}$ is the average number of visitors per car.

We have assumed that v_{total} is 30,000 at peak, p_{car} is about 60%, and $v_{per\ car}$ is 3. Therefore, we can calculate that the approximate IVC at peak is 6,000 cars.

Next, we will calculate the total area the cars should occupy. We designed $6\mathrm{m} \times 4\mathrm{m}$ parking spaces, which provide ample space for the cars. Therefore, the total area of the parking spaces should be:

$$6{,}000\ \times 6m\ \times 4m = 144{,}000m^2$$

Adding the total area of the roads and the gaps between the cars, we reasoned that $150{,}000m^2$ of land will be enough for most of the cars when the volume of visitors is at peak.

2) A square for major activities

We will put a central square in the center of our zoo for major activities. We included a lake in the central square for *recreation* (visitors can take lake tours and enjoy the view). The area of the central square and the lake is quite flexible, and they can change according to the space and shape of other elements.

3) Restaurants & stands for eating

We designed a huge in-lake restaurant in the lake of the central square, also for visitors' *recreation*. The in-lake restaurant is the main area for eating, and our zoo will also offer ice creams, hot dogs, and barbecue skewers at varies mobile stands evenly distributed in the entire space. Since the stands are mobile, we will not include them in our map.

4) Roads wide enough for visitors to move around freely

We researched that the normal width of main roads in a big park should be at least 5 meters wide to avoid congestion. Therefore, we will include roads more than 5 meters wide in *Veridian Ark*. We decided to design **biome 2** (Temperate Grassland biome) as a Safari Ride for visitors. The other biomes will each include about 20 exhibitions displaying different species. In order to achieve wildlife conservation, we should also ensure that each individual species has enough space to live and move around.

So far, our map looks like this:

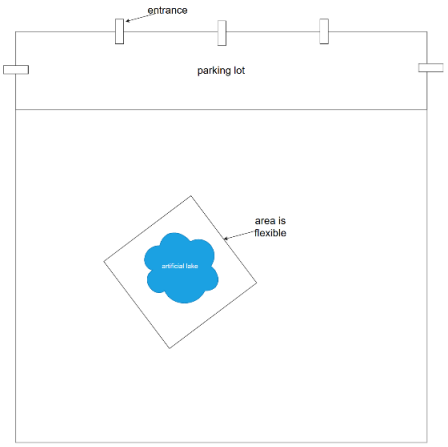


Figure 2.3.1 First Draft of Map

We made sure that the parking lot occupies an entire edge of the biggest square, so that visitors' cars can come in from every direction, and it won't impact much on the shapes of other areas.

Then, we used the rest space to design the place and shape of the five biomes and the aquarium:

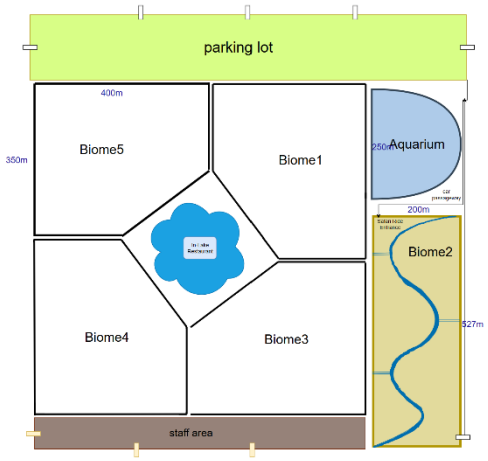


Figure 2.3.2 Second Draft of Map

In this design, **biome 1**, **biome 3**, **biome 4**, and **biome 5** all have areas of 125,000m^2, and **biome 2**, used for safari rides, has an area of 105,400m^2.

The aquarium is designed with an arc-shaped edge, so that there will be enough space for safari cars to pass. Therefore, it makes the area of the aquarium a little less than 50,000m^2.

There is also a huge area left for the staff (about 54,576m^2), so that they can rest and hold major meetings there.

At last, we designed the interior appearance of the five biomes, shown in **Figure 2.3.3**.



Figure 2.3.3 Third Draft of Map

For the four biomes in the center, there are two main roads running through the ends of each one. The main roads help connect the main entrances of the biomes. For the interior design of the four biomes, considering that there may be predator and prey relationships, we should keep enough space between different exhibitions. The buffers between different exhibitions (also the passageway for visitors) are about five meters wide. As stated previously, we kept animals of various classes and sizes to ensure animal diversity. Species' living places depend on two factors:

1) As stated in 2.2, we developed a ranking system to represent the size of an animal species. The animals with the ranking of 3 get bigger exhibition spaces, while animals with the ranking of 1 get relatively smaller ones.

2) If an animal species needs big spaces to move around, they get bigger parts. In addition, there will be individual interaction areas, gift shops, washrooms, and various mobile food stands (not presented in the map) in each of the four biomes. Also, employees are given ample space to rest in the green staff's areas.

Moreover, animals with higher scarcity are put in more central parts of a biome, since they have higher scientific research and education value.

For **biome 2**, we let the animals roam around but also provided them with an approximate living space. In order to ensure the safety of the animals, aggressive predators, like lions, are kept alone in cages large enough for them to roam around freely. The entire ride, a little less than 1 kilometer, offers visitors an excellent view of the ecological environment with the animals in their most natural states.

For a clearly labeled scale drawing of *Veridian Ark*, see 3.1.

3. Solution

3.1 Labeled Scale Drawing of Zoo Design

Our labeled scale drawing of *Veridian Ark* is shown in **Figure 3.1.1** below.

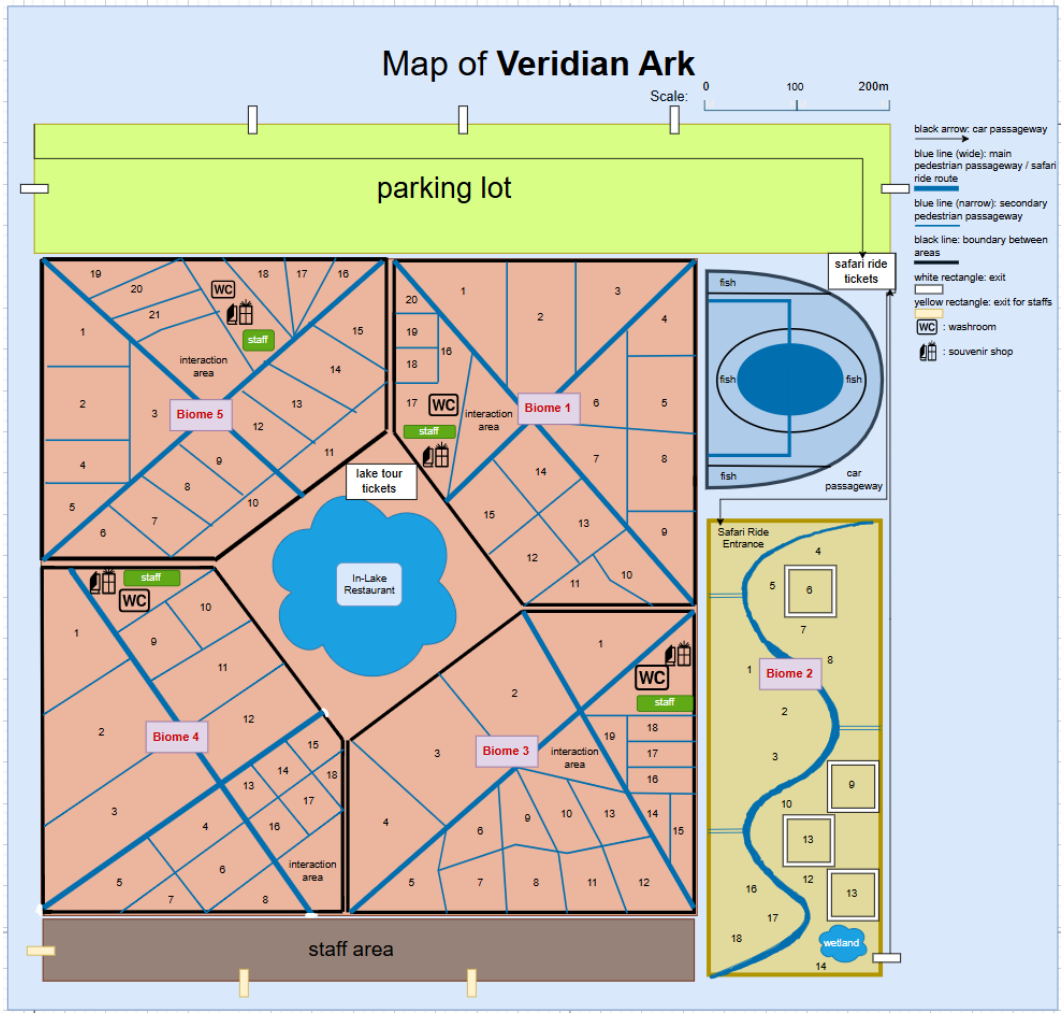


Figure 3.1.1 Map of *Veridian Ark*

In this final draft of our map, we included a map scale on the top-right corner, and gave each area a clearly-labeled biome number. Below are the species the numbers in each biome stand for.

Table 3.1.1 Animals in **Biome 1**

Number	Common Name	Technical Name	Number	Common Name	Technical Name
1	Bornean Orangutan	*Pongo pygmaeus*	11	Matschie's Tree-kangaroo	*Dendrolagus matschiei*
2	Asian Elephant	*Elephas maximus*	12	Lar Gibbon	*Hylobates lar*

3	South American Tapir	*Tapirus terrestris*	13	Hanuman Langur	*Presbytis entellus*
4	King Cobra	*Ophiophagus hannah*	14	Rhinoceros Hornbill	*Buceros rhinoceros*
5	Brown-throated Sloth	*Bradypus variegatus*	15	African Grey Parrot	*Psittacus erithacus*
6	Scarlet Macaw	*Ara macao*	16	Dyeing Poison Dart Frog	*Dendrobates tinctorius*
7	Hyacinth Macaw	*Anodorhynchus hyacinthinus*	17	Oriental Garden Lizard	*Calotes versicolor*
8	Giant Anteater	*Myrmecophaga tridactyla*	18	Stick Insects	*Phasmatodea spp.*
9	Red Bird-of-Paradise	*Paradisaea rubra*	19	Peacock Mantis Shrimp	*Odontodactylus scyllarus*
10	Fer-de-lance	*Bothrops asper*	20	Jamaican Mango Hummingbird	*Trochilus polytmus*

Table 3.1.2 Animals in **Biome 2**

Number	**Common Name**	**Technical Name**	**Number**	**Common Name**	**Technical Name**
1	Black Rhinoceros	*Diceros bicornis*	10	Golden Pheasant	*Chrysolophus pictus*
2	Giraffe	*Giraffa camelopardalis*	11	Lion	*Panthera leo*
3	African Bush Elephant	*Loxodonta africana*	12	African Wild Ass	*Equus africanus*
4	Plains Zebra	*Equus quagga*	13	African Rock Python	*Python sebae*
5	Thomson's Gazelle	*Eudorcas thomsonii*	14	Greater Flamingo	*Phoenicopterus roseus*
6	Cheetah	*Acinonyx jubatus*	15	Round-eared Elephant Shrew	*Macroscelides proboscideus*
7	Springbok	*Antidorcas marsupialis*	16	Glossy Ibis	*Plegadis falcinellus*
8	Common Ostrich	*Struthio camelus*	17	Checked Elephant Shrew	*Rhynchocyon cirnei*
9	African Wild Dog	*Lycaon pictus*	18	Cape Hare	*Lepus capensis*

Table 3.1.3 Animals in **Biome 3**

Number	**Common Name**	**Technical Name**	**Number**	**Common Name**	**Technical Name**
1	Boa Constrictor	*Boa constrictor*	11	Egyptian Spiny-tailed Lizard	*Uromastyx aegyptia*
2	Arabian Oryx	*Oryx leucoryx*	12	Black-tailed Jackrabbit	*Lepus californicus*
3	Tibetan Antelope	*Pantholops hodgsonii*	13	Desert Tortoise	*Gopherus agassizii*
4	Dromedary Camel	*Camelus dromedarius*	14	Fennec Fox	*Vulpes zerda*
5	Coachwhip Snake	*Masticophis flagellum*	15	Lesser Egyptian Jerboa	*Jaculus jaculus*
6	Great Horned Owl	*Bubo virginianus*	16	Plains Pocket Gopher	*Geomys bursarius*
7	Isabelline Shrike	*Lanius isabellinus*	17	Red-spotted Toad	*Bufo punctatus*
8	Greater Roadrunner	*Geococcyx californianus*	18	Mongolian Gerbil	*Meriones unguiculatus*
9	Saker Falcon	*Falco cherrug*	19	Kit Fox	*Vulpes macrotis*
10	Sand Cat	*Felis margarita*			

Table 3.1.4 Animals in **Biome 4**

Number	Common Name	Technical Name	Number	Common Name	Technical Name
1	Steppe Eagle	*Aquila nipalensis*	10	Bobak Marmot	*Marmota bobak*
2	Gray Wolf	*Canis lupus*	11	Red-tailed Hawk	*Buteo jamaicensis*
3	Pronghorn	*Antilocapra americana*	12	Peregrine Falcon	*Falco peregrinus*
4	White-tailed Jackrabbit	*Lepus townsendii*	13	Calandra Lark	*Melanocorypha calandra*
5	Pallid Harrier	*Circus macrourus*	14	Black-tailed Prairie Dog	*Cynomys ludovicianus*
6	Red Fox	*Vulpes vulpes*	15	Gunnison's Prairie Dog	*Cynomys gunnisoni*
7	Killdeer	*Charadrius vociferus*	16	Eastern Fence Lizard	*Sceloporus undulatus*
8	Coyote	*Canis latrans*	17	Meadow Vole	*Microtus pennsylvanicus*
9	Eastern Diamondback Rattlesnake	*Crotalus adamanteus*	18	Chinese Zokor	*Myospalax fontanieri*

Table 3.1.5 Animals in **Biome 5**

Number	Common Name	Technical Name	Number	Common Name	Technical Name
1	Moose	*Alces alces*	12	Snowy Owl	*Bubo scandiacus*
2	Elk	*Cervus canadensis*	13	Great Grey Owl	*Strix nebulosa*
3	Mountain Goat	*Oreamnos americanus*	14	Western Marsh Harrier	*Circus aeruginosus*
4	Canada Lynx	*Lynx canadensis*	15	Tawny Owl	*Strix aluco*
5	Snowshoe Hare	*Lepus americanus*	16	American Red Squirrel	*Tamiasciurus hudsonicus*
6	Wolverine	*Gulo gulo*	17	Eastern Chipmunk	*Tamias striatus*
7	Sable	*Martes zibellina*	18	Ferret	*Mustela putorius furo*
8	Arctic Fox	*Vulpes lagopus*	19	Black-backed Woodpecker	*Picoides arcticus*
9	Pine Marten	*Martes martes*	20	Common Frog	*Rana temporaria*
10	North American River Otter	*Lontra canadensis*	21	Common Viper	*Vipera berus*
11	Black-billed Capercaillie	*Tetrao parvirostris*			

3.2 Visitor’s Guide

The following two pages are the visitor’s guide that we will hand to the visitors at the entrance of the zoo.



Biome 1
1. Bornean Orangutan (Pongo pygmaeus)
2. Asian Elephant (Elephas maximus)
3. South American Tapir (Tapirus terrestris)
4. King Cobra (Ophiophagus hannah)
5. Brown-throated Sloth (Bradypus variegatus)
6. Scarlet Macaw (Ara macao)
7. Hyacinth Macaw (Anodorhynchus hyacinthinus)
8. Giant Anteater (Myrmecophaga tridactyla)
9. Red Bird-of-Paradise (Paradisaea rubra)
10. Fer-de-lance (Bothrops asper)
11. Matschie's Tree-kangaroo (Dendrolagus matschiei)
12. Lar Gibbon (Hylobates lar)
13. Hanuman Langur (Presbytis entellus)
14. Rhinoceros Hornbill (Buceros rhinoceros)
15. African Grey Parrot (Psittacus erithacus)
16. Dyeing Poison Dart Frog (Dendrobates tinctorius)
17. Oriental Garden Lizard (Calotes versicolor)
18. Stick Insects (Phasmatodea spp.)
19. Peacock Mantis Shrimp (Odontodactylus scyllarus)
20. Jamaican Mango Hummingbird (Trochilus polytmus)

Biome 2
1. Black Rhinoceros (Diceros bicornis)
2. Giraffe (Giraffa camelopardalis)
3. African Bush Elephant (Loxodonta africana)
4. Plains Zebra (Equus quagga)
5. Thomson's Gazelle (Eudorcas thomsonii)
6. Cheetah (Acinonyx jubatus)
7. Springbok (Antidorcas marsupialis)
8. Common Ostrich (Struthio camelus)
9. African Wild Dog (Lycaon pictus)
10. Golden Pheasant (Chrysolophus pictus)
11. Lion (Panthera leo)
12. African Wild Ass (Equus africanus)
13. African Rock Python (Python sebae)
14. Greater Flamingo (Phoenicopterus roseus)
15. Round-eared Elephant Shrew (Macroscelides proboscideus)
16. Glossy Ibis (Plegadis falcinellus)
17. Checked Elephant Shrew (Rhynchocyon cirnei)
18. Cape Hare (Lepus capensis)

Biome 3
1. Boa Constrictor (Boa constrictor)
2. Arabian Oryx (Oryx leucoryx)
3. Tibetan Antelope (Pantholops hodgsonii)
4. Dromedary Camel (Camelus dromedarius)
5. Coachwhip Snake (Masticophis flagellum)
6. Great Horned Owl (Bubo virginianus)
7. Isabelline Shrike (Lanius isabellinus)
8. Greater Roadrunner (Geococcyx californianus)
9. Saker Falcon (Falco cherrug)
10. Sand Cat (Felis margarita)
11. Egyptian Spiny-tailed Lizard (Uromastyx aegyptia)
12. Black-tailed Jackrabbit (Lepus californicus)
13. Desert Tortoise (Gopherus agassizii)
14. Fennec Fox (Vulpes zerda)
15. Lesser Egyptian Jerboa (Jaculus jaculus)
16. Plains Pocket Gopher (Geomys bursarius)
17. Red-spotted Toad (Bufo punctatus)
18. Mongolian Gerbil (Meriones unguiculatus)
19. Kit Fox (Vulpes macrotis)

Biome 4
1. Steppe Eagle (Aquila nipalensis)
2. Gray Wolf (Canis lupus)
3. Pronghorn (Antilocapra americana)
4. White-tailed Jackrabbit (Lepus townsendii)
5. Pallid Harrier (Circus macrourus)
6. Red Fox (Vulpes vulpes)
7. Killdeer (Charadrius vociferus)
8. Coyote (Canis latrans)
9. Eastern Diamondback Rattlesnake (Crotalus adamanteus)
10. Bobak Marmot (Marmota bobak)
11. Red-tailed Hawk (Buteo jamaicensis)
12. Peregrine Falcon (Falco peregrinus)
13. Calandra Lark (Melanocorypha calandra)
14. Black-tailed Prairie Dog (Cynomys ludovicianus)
15. Gunnison's Prairie Dog (Cynomys gunnisoni)
16. Eastern Fence Lizard (Sceloporus undulatus)
17. Meadow Vole (Microtus pennsylvanicus)
18. Chinese Zokor (Myospalax fontanieri)

Biome 5
1. Moose (Alces alces)
2. Elk (Cervus canadensis)
3. Mountain Goat (Oreamnos americanus)
4. Canada Lynx (Lynx canadensis)
5. Snowshoe Hare (Lepus americanus)
6. Wolverine (Gulo gulo)
7. Sable (Martes zibellina)
8. Arctic Fox (Vulpes lagopus)
9. Pine Marten (Martes martes)
10. North American River Otter (Lontra canadensis)
11. Black-billed Capercaillie (Tetrao parvirostris)
12. Snowy Owl (Bubo scandiacus)
13. Great Grey Owl (Strix nebulosa)
14. Western Marsh Harrier (Circus aeruginosus)
15. Tawny Owl (Strix aluco)
16. American Red Squirrel (Tamiasciurus hudsonicus)
17. Eastern Chipmunk (Tamias striatus)
18. Ferret (Mustela putorius furo)
19. Black-backed Woodpecker (Picoides arcticus)
20. Common Frog (Rana temporaria)
21. Common Viper (Vipera berus)

Welcome to *Veridian Ark*!

Explore about 100 species from around the world, enjoy interactive animal encounters, and learn about the magic of wildlife with our family-friendly guides.

Five biomes of animals, an aquarium, and a lake is involved in our zoo. The topics of these five biomes are:

- **Biome 1** Tropical Rainforest
- **Biome 2** Savanna
- **Biome 3** Desert / Arid
- **Biome 4** Temperate Grassland / Prairie
- **Biome 5** Boreal Forest / Taiga

Each has its unique style and feature. Feel free to discover each area on your own, but a rational tour route may help you maximize your experience in the zoo.

Two routes are suggested:

1. Tour route: Biome 1 → Aquarium → Biome 2 →
   Biome 3 → Lake → Biome 4 → Biome 5
   Suitable for: people driving cars / entering the zoo in an entrance in Parking Lot
2. Tour route: Biome 2 → Aquarium → Biome 1 →
   Biome 5 → Lake → Biome 4 → Biome 3
   Suitable for: people entering the zoo in the entrance connected with Biome 2

Special Features of the zoo:

- Safari Ride in Biome 2
- In-Lake Restaurant
- Thematic Souvenir Shops in Each Biome

Have a good day in *Veridian Ark*!

3.3 Memo for Zoo Designers

Memo for Zoo Designers over the World

To: zoo designers over the world

From: *Veridian Ark* designers

The main requirement for zoo designers is to design a zoo that can contribute to animal conservation, ensuring visitor experience and provide a foundational environment for scientific research. Every zoo designer needs to have the following ability, to create a well-designed zoological park that could be used for a different collection of animals:

- Knowing the basic knowledge of animal species and biomes, this can give a better structure for animal living. We firstly researched for every animal species (150 species) that we may choose, for their living condition, relationship with other animals, etc. We have different size of cages or living area covering a big range of animals. So, the zoo is also more flexible if other animal species are moving in.
- Deeply consider for animals that are contained in the zoo. Every different animal species is suitable for different living conditions. Design an appropriate living area for animal can ensure their well-being. Our zoo provides living areas for animal species that resembles to their original living environment. Our zoo has professional staff that can take care for animals in each biome section, they can help animals that are in need or injured, so the animals' healthy living will be guaranteed.
- Deeply consider for visitors. Visitors will go to zoo for education and entertainment, the structure for the zoo should be friendly for visitors (i.e., no long-distance walk) Visitor expects high quality zoo structure and road arrangement, so we as designers should review our zoo from a visitor perspective.
- Balancing of the zoo structure, each biome should contain the same amounts of animals, giving a balanced structure. The even distribution of living area for animal species can give animals better feelings in a zoo that is unfamiliar for them. Especially for grass biome, even though they are all living on one big grassland, the grassland is big enough for them to live apart, not disturbing each other, so every animal will still be safe if another new animal joins this biome section. It could also be easier for visitor to find the animal they want to see.
- A well-designed zoo should not only have animals, where restaurant and interacting areas should also be included, giving visitors satisfaction and relaxing space
- Creativity: creativity is vital for a well-designed zoo, because visitors usually expect surprise. A zoo with more creation can give visitor more special memories. The shape biome sections in our zoo refer to the Pythagorean theorem chord diagram.

3.4 Letter for COMAP

Dear COMAP:

We are honored to present our design for a 1-square-kilometer zoo, named **Veridian Ark**, created to fulfill the four essential purposes: 1) *wildlife conservation*, 2) *scientific research*, 3) *education*, and 4) *recreation*. The name reflects our vision of a green sanctuary where every species is happy and healthy.

Below, we summarize what we did addressing the four primary purposes:

1) *Wildlife conservation:* To achieve *wildlife conservation*, every species is provided with enough living space to ensure its well-being. We also included the buffers - at least five meters wide - between predators and prey in order to guarantee the animals' safety.
2) *Scientific research:* To achieve *scientific research*, we referred to data from IUCN's criteria to determine the animals' scarcity. The animals of higher scarcity have higher scientific research value, and thereby are more likely to be chosen for our animal collection.
3) *Education:* To achieve *education*, we need to include species that have distinctive traits or behaviors. Therefore, we gave them higher education value, resulting in a higher possibility of being chosen.
4) *Recreation:* To achieve recreation, we assessed how different animals are perceived by students and adults. For instance, while lions might be intimidating to students, they often appear majestic and captivating in an adult's eyes. Hence, animals that are likely to be accepted by both students and adults have high recreation value.

The layout of Veridian Ark is mainly divided into 5 different parts:

a) Five biomes all simulating the living conditions of the wild:
   Biome1: Tropical Rainforest Biome
   Biome2: Savanna Biome
   Biome3: Desert / Arid Biome
   Biome4: Temperate Grassland / Prairie Biome
   Biome5: Boreal Forest / Taiga Biome)
b) An aquarium displaying fish species
c) A central square for major activities
d) A giant parking lot
e) Areas for the staff to rest and hold meetings

Specifically, each of the five biomes contains about 20 species. For the exhibitions, we made sure that every species has enough living space to thrive and move around freely. Other sections, like c), are aimed towards increasing the visitors' *recreation*.

In conclusion, our zoo, Veridian Ark, is designed not only to serve the visitors, but also to offer a place where wildlife species are protected and allowed to thrive.

Sincerely,
Team 16970

4. Analysis & Conclusion

4.1 Implication & Analysis of Assumptions

Our model not only designs a zoo, but also give zoo designing a digitalized method. This method can be also used to design a zoo with a different combination of animal species, as long as they can be divided into several biomes and their size is relatively average. However, if some assumptions change, some of our zoo design might be different. The following table discusses situations where one or some the assumptions listed in **1.2** changes.

Table 4.1.1 Changes of Assumptions

Changes of Assumptions	Changes of Zoo Design
The pedestrian volume increases / decreases.	The parking lot need to be bigger, so the area left for animals decreases, meaning that we need to involve less species, vice versa.
Souvenir shops, restaurants, and a lake are not included in the zoo.	Although we can include more species, the *recreation* value will decrease significantly.
The ratio between student visitors and adult visitors is not 2:3.	The data in the evaluation model will change directly, resulting in a different selection of species. For instance, if the ratio is 1:1, the weight of *student recreation* will become higher, thus aggressive animals will have a lower rank because of its low score on *student recreation*.
The shape of the zoo is a circle.	Since the center of the circle has the same distance to each spot on the circumference, visitors will not be likely to walk long distances. However, as stated in **1.2**, visitors will be easy to get lost.

4.2 Strengths & Limitations

Strengths:

- In our model, all four of the main purposes of zoos are carefully considered.
- Different sections of *Veridian Ark* are presented clearly on our map.
- Our selection of animals is done carefully with two steps, using Excel and Python.
- We give different animals different living areas based on their size and scarcity.

Limitations and improvements:

- Limitation: Our animal selection process is mostly done by hand. Though our model can be used for another combination of species, significant workload will be involved.
  Improvement: Use more formulae and reach for a more automated process.
- Limitation: Some of our data in the evaluation model is from our subjective view. This might cause rankings to be inaccurate for different people.
  Improvement: Do more surveys and learn about most people's favors.
- Limitation: The shape of the map is defaulted to be a square. Though we thought about other shapes, we didn't use data to calculate which shape is the best
  Improvement: Construct another evaluation model. Try different shapes and calculate their score using data to reveal the best shape.
- Limitation: Our zoo is big in size; people need to spend a lot of time on walking long distances.
  Improvement: Set a sightseeing vehicle that can help visitors travel from one place to another quickly.

4.3 Conclusion

Above are all of our results. In **Section 1**, we briefly explained the problem, our interpretations, and some assumptions we made while designing the zoo. In **Section 2**, we explained the elaborate process of zoo designing: used Excel to get an initial ranking of species, used Python to complete the final selection of animals, and drew the map based on the size and scarcity of the species. In **Section 3**, we presented our final results of our zoo design. In **Section 4**, we reflected on our process: discussed situations when assumptions changed and analyzed strengths and limitations in our design. In future study, we will try to solve some of our limitations, reach for a more comprehensive model, and keep improving our design to practically match the visitors' demands and the four main purposes of zoos.

Reference

1. https://www.drawio.com/
2. https://www.iucn.org/

Usage of AI

AI tool: https://www.doubao.com/

1. Input: Give me 30 common species of each biome (tropical rainforest, savanna, desert / arid, temperate grassland, prairie, and boreal forest / taiga) and give me their technical name.
   Output: (The 150 species shown in **Figure 2.1.1)**
2. Input: Give me an image of each of these species.
   Output: (150 images)